Module Four Assignment

Filip Arghir

Southern New Hampshire University

CS 499 Computer Science Capstone

Wasim Alim

5/27/2026

Module Four Assignment

For Milestone Three, I continued with my CS 330 Computational Graphics and Visualization final project as my artifact for the Algorithms and Data Structures category. The scene still includes the two cars, the lamp post, the sun, the tree, the grass island, textures, and lighting, as well as materials and camera interaction. The changes made to the artifact did alter the lighting, which I am very excited about, and this project highly connects with my career goals in graphics programming and my career choice of being a technical artist.

I decided to continue working on this artifact and add it to my ePortfolio since it shows visual and technical problem-solving. The project is not meant to be just a simple program that shows an output of objects. It is a visual scene where the code directly controls what appears on screen. This makes it a great artifact for showing my ability to work with rendering logic, materials, transformations, textures, and 3D scenes. For this milestone, I wanted to continue improving this project, specifically from an algorithmic and data structure perspective, and by making part of the rendering system more organized as well as easier to expand.

Before my enhancement, I added a scene object struct to store information that needs to be rendered into a single static object. This is a mesh type, position, scale, rotation, texture tag, material tag, color, and UV scale. I also added a vector called m_sceneobjects to store the static objects in one organized list. I made a buildsceneobjectlist function to fill that vector and then made a function to render each of those objects based on the information stored in the struct. My renderScene function has static objects that are rendered through a loop instead of being called manually one by one.

Some other changes were with a meshtype enum so each object can identify which type of mesh it uses, meaning a plane, cylinder, cone, torus, sphere, or box. This made the render

helper much easier to understand because it can use a switch statement to decide which mesh draw function to call. I kept the vehicle rendering functions separate since each car is made from multiple pieces with different transformations, textures, and materials. Trying to force the vehicles into one simple scene object would cause more complications and messier code instead of a cleaner project.

This enhancement improved the artifact because it separates some of the scene data from rendering behavior. Instead of every simple object being rendered through a repeated function call, the static scene objects are stored in a data structure and then processed through a loop. This makes the project easier to expand because new, simple objects can be added to the vector rather than requiring a new render function every time. It also demonstrates my ability to use data structures to solve organizational problems, as well as scalability problems in my graphics project.

The main course outcome that this enhancement supports is the ability to design and evaluate computing solutions using algorithmic principles as well as data structures while managing design trade-offs. I used a struct, vector, and enum, as well as a loop, to make the rendering system more organized. I also had to decide which objects fit this system and which ones should stay separate. That trade-off was important because the static single mesh objects fit well into the vector system, while the vehicles were too complex and were better left as just dedicated rendering functions.

These enhancements also support the outcome related to using computing tools and techniques to implement solutions that have value. The project runs inside the original provided SNHU project that uses an OpenGL framework, but is now a part of the rendering system that is more scalable, easier to maintain, and handle. It supports professional communication as well

because I can explain the difference between the original manual rendering approach and the enhanced data-driven rendering approach.

The biggest challenge I faced was deciding how far to go with the enhancements. I did not want to turn my project into a full game engine as the scope would be too much for me, nor did I want to completely rewrite the scene. I wanted the changes to be meaningful but still realistic for the milestone. At first, it was tempting to move every object into the new structure system, but the vehicles are made of multiple meshes and would have required a much more complicated structure. I decided to keep the vehicles separate and, therefore, focus on the simpler static objects. This made the enhancements cleaner and easier to explain and execute.

This milestone allowed me to better understand how data structures can help improve a project without changing what the user sees visually. The scene still looks the same, feels the same, but the way the objects are organized and rendered is better. This showed me that algorithms as well as data structures are not only about sorting or searching, but that they can also be used to organize real project data and then make software easier to expand later.